

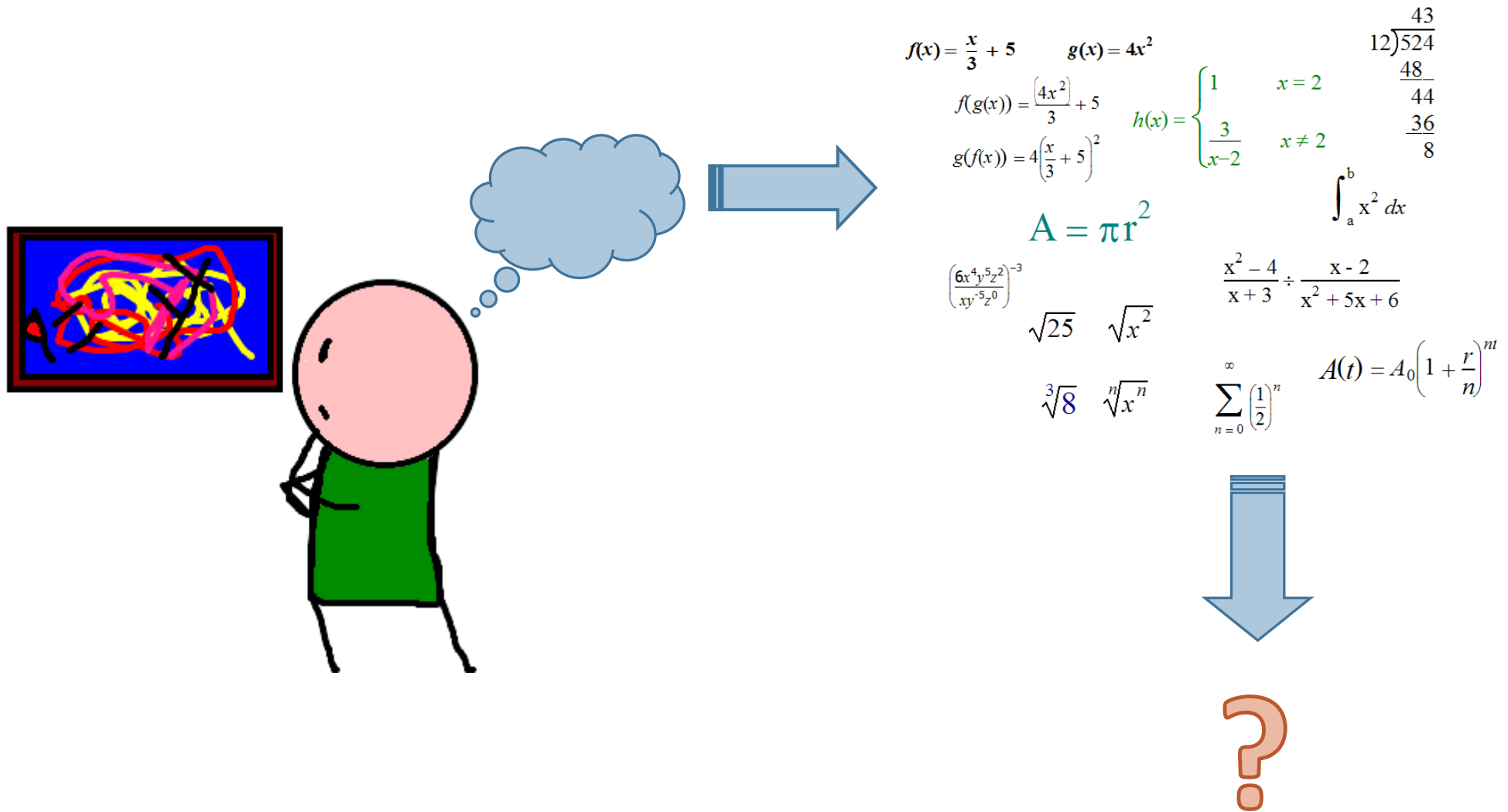
Mathematical Modeling

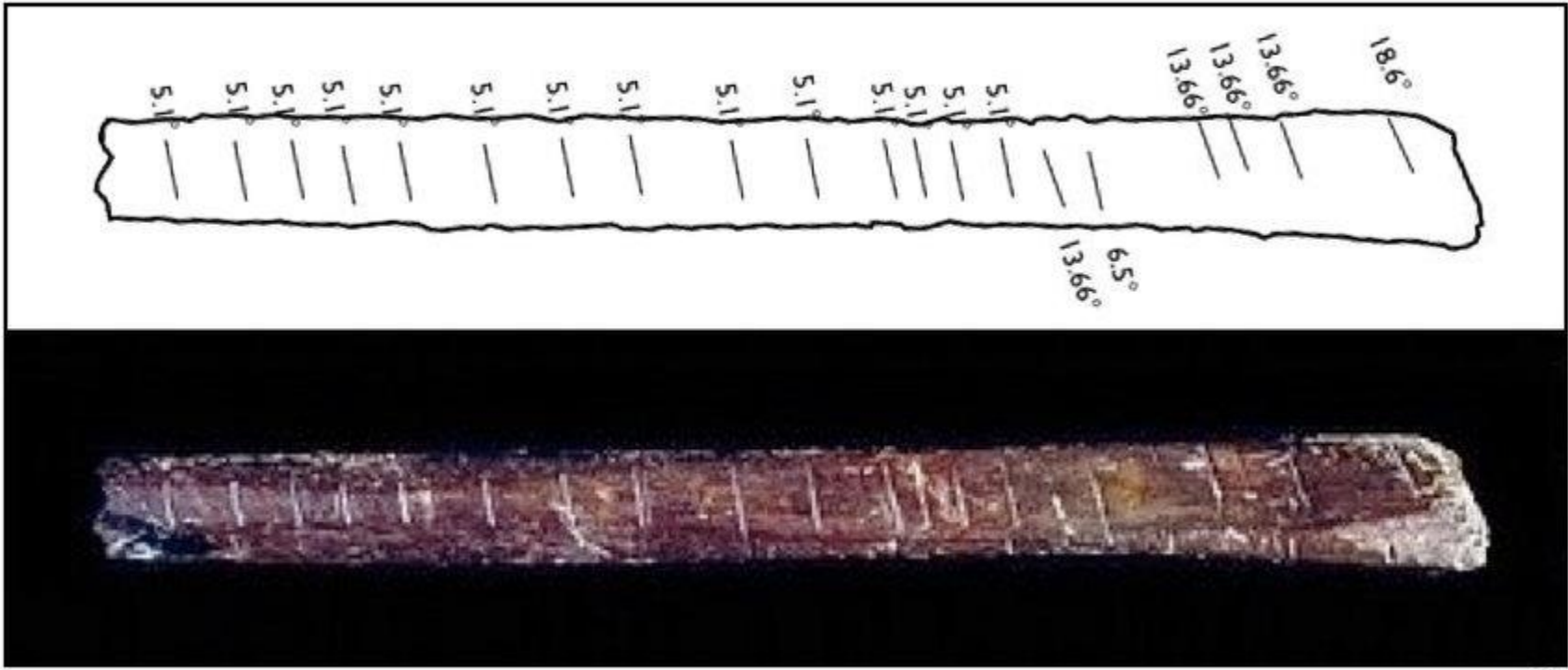
PATRICIO ORIO

CINV – UV

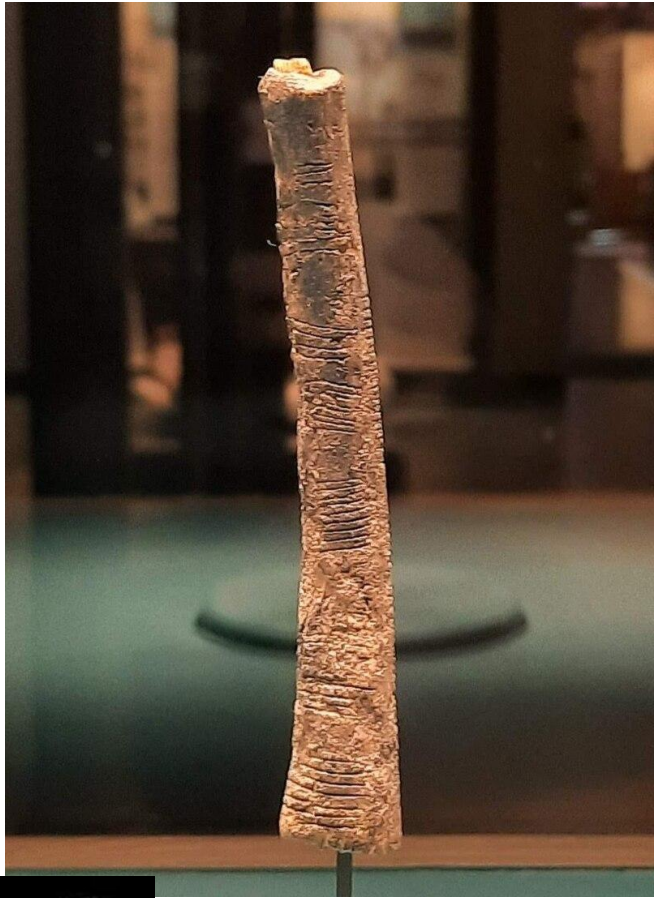
2026

¿What is a mathematical model?

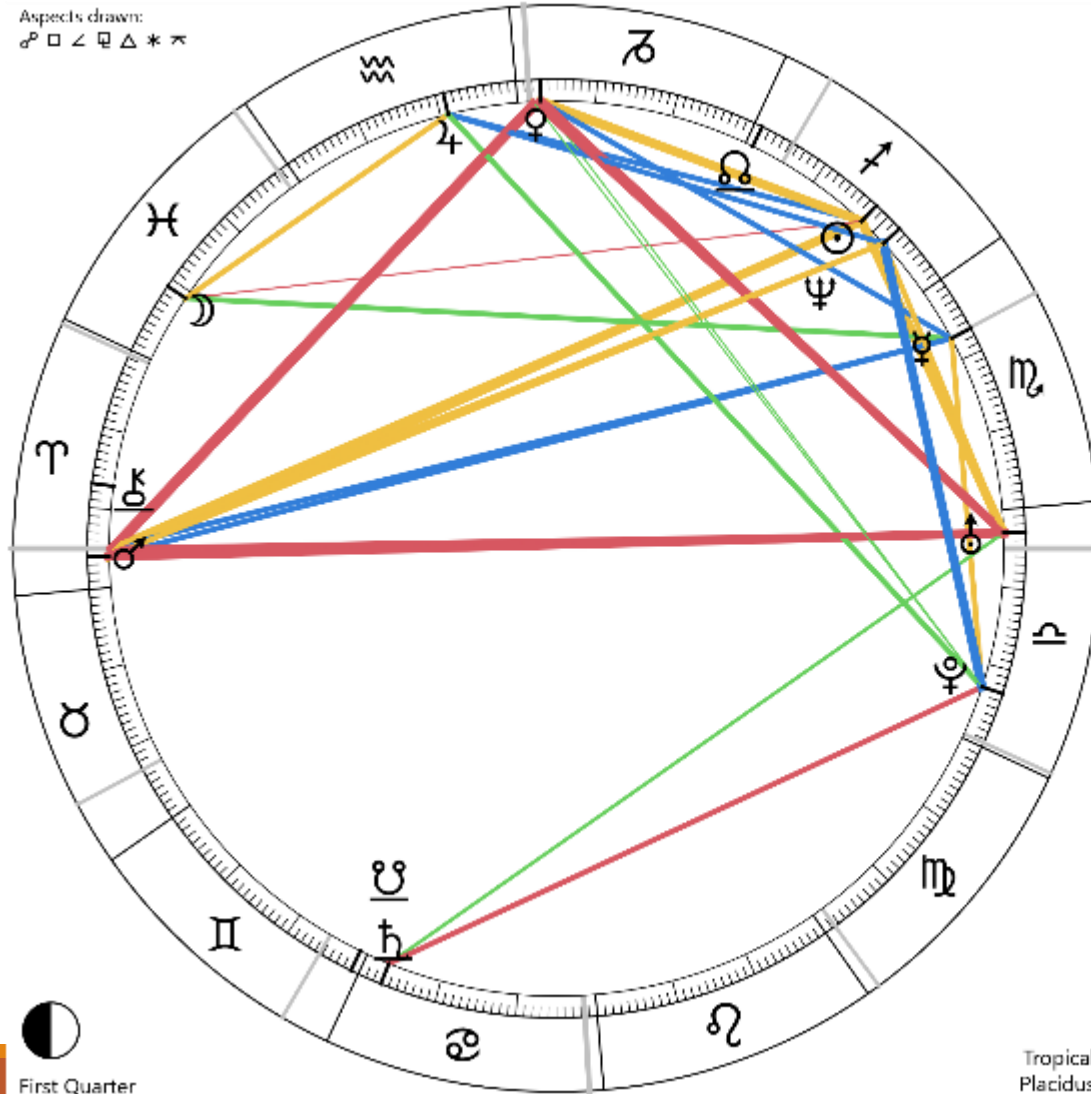




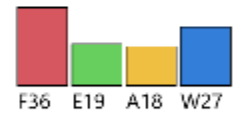
5



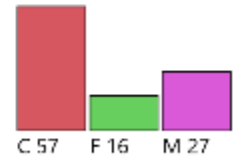
A repeating pattern of zodiac signs on a dark background. The signs are depicted in a stylized, line-art manner, including Aries (ram), Taurus (bull), Gemini (twins), Cancer (crab), Leo (lion), Virgo (maiden), Libra (scales), Scorpio (scorpion), Sagittarius (archer), Capricorn (goat), Aquarius (water bearer), and Pisces (fish). The pattern is dense and covers the entire surface.



Elements



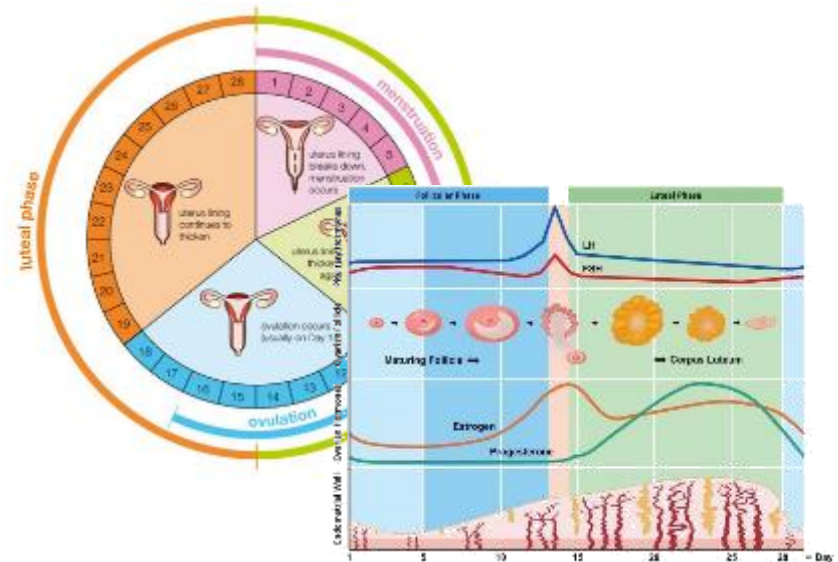
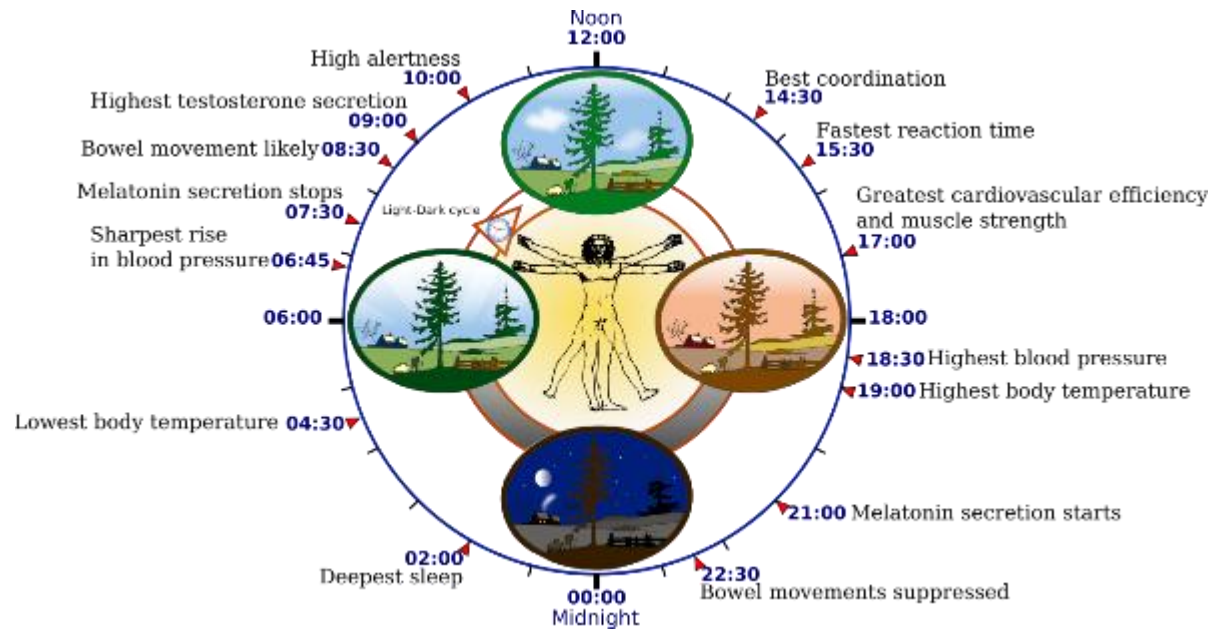
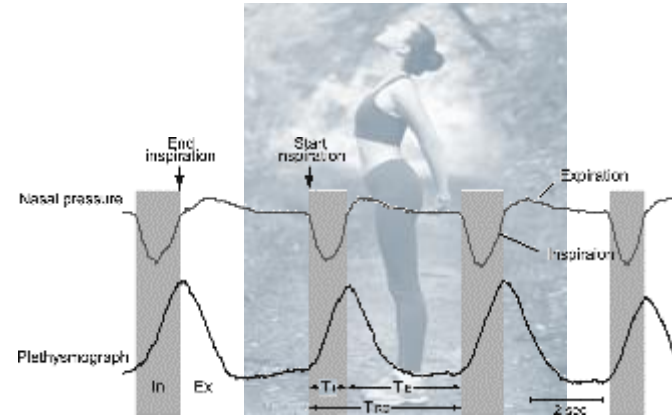
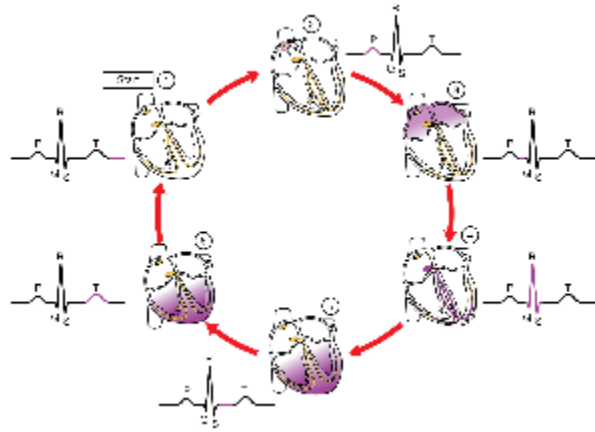
Qualities



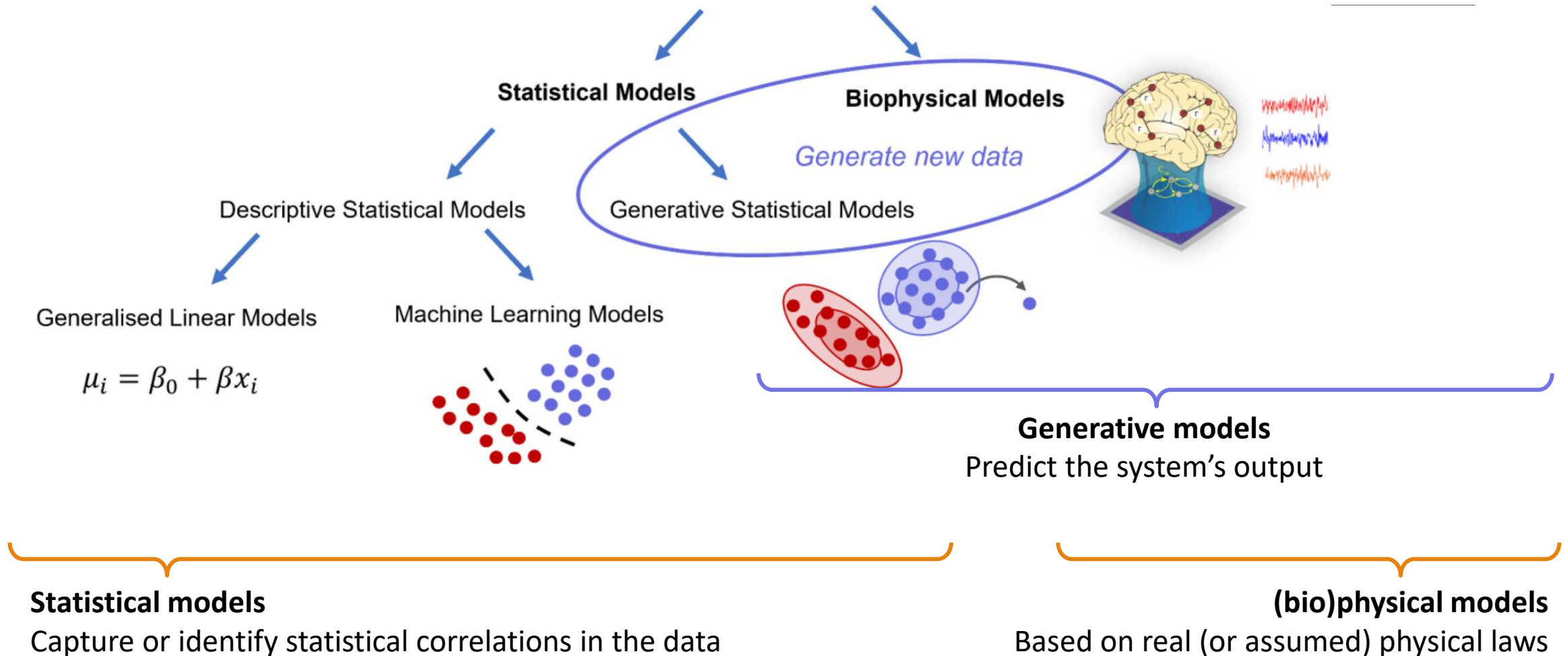
AC	24 ar 34
2. House	22 ta 40
3. House	24 ge 14
MC	28 cp 03
11. House	0 pi 59
12. House	0 ar 08

Tropical
Placidus

Biological rhythms



Types of models

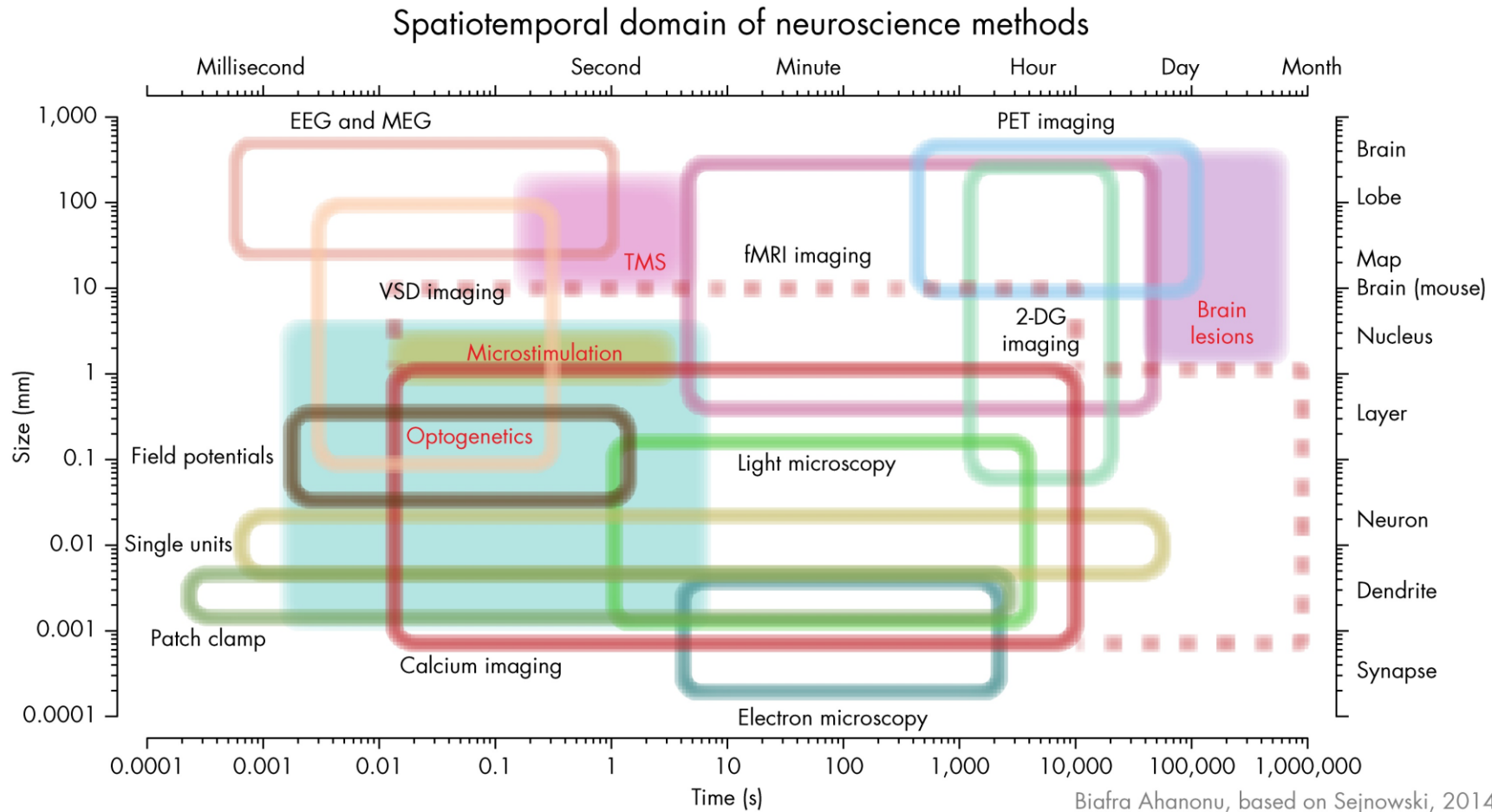


Why model?

- We study the brain with many different tools



- Molecular / Cellular Biology
- Anatomy
- Imaging – structure
- Imaging – function (fMRI)
- Electrophysiology (EEG)
- Psychology and Behavior

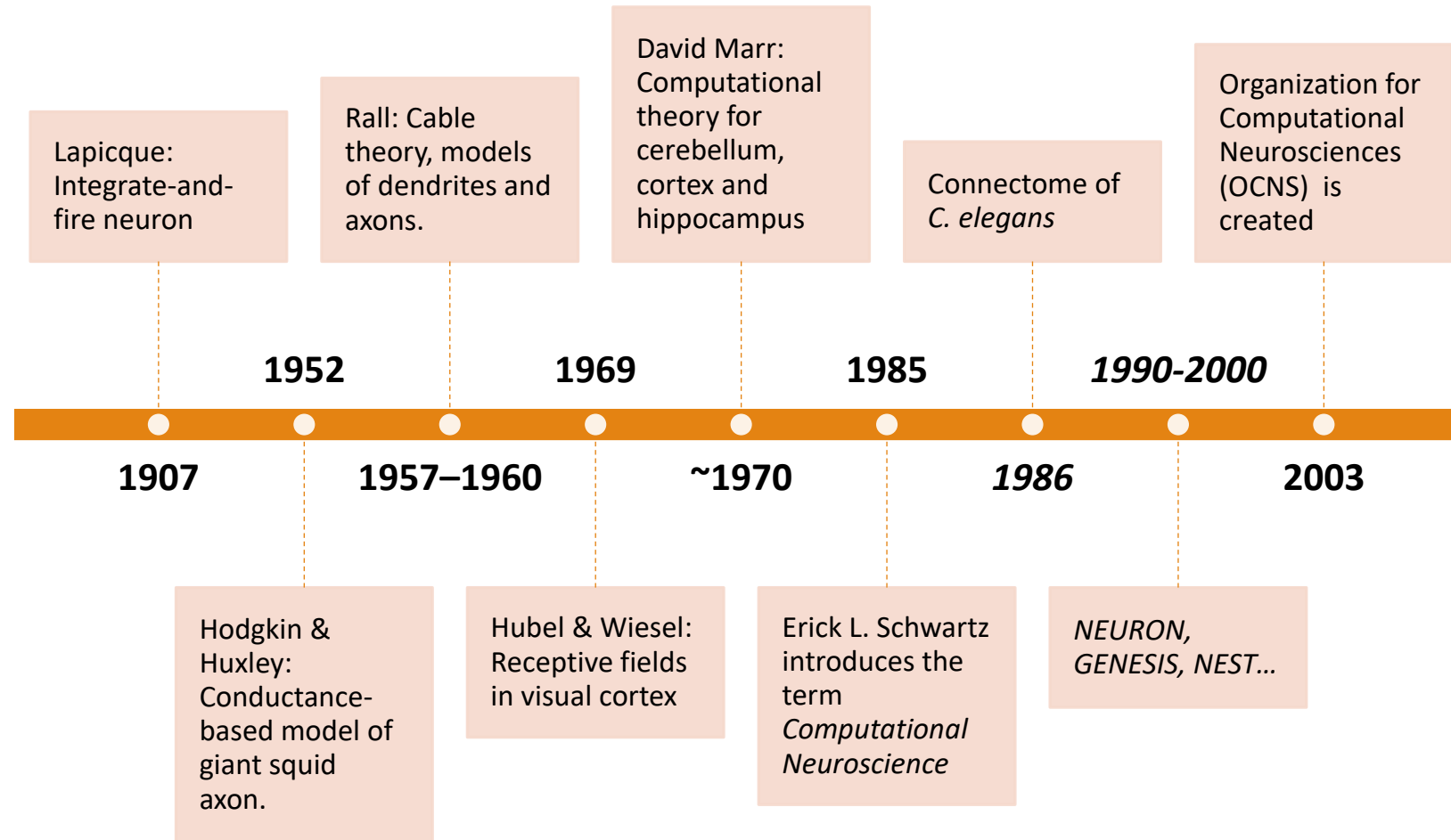


Mathematical modeling in neuroscience



Figure 1.8: Neurons are dynamical systems.

Izhikevich, E. (2007)

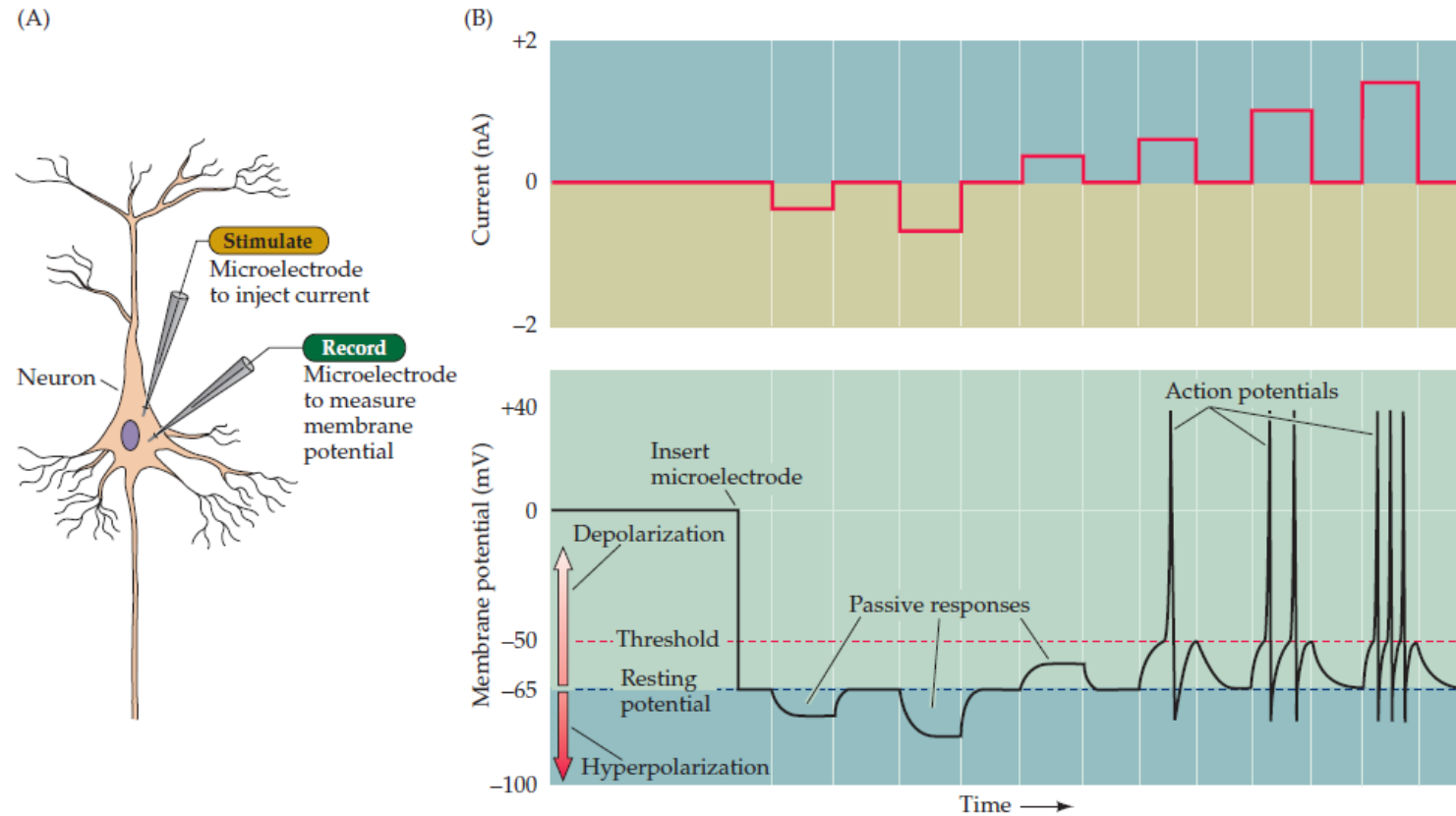


Tools for modelers

- The Virtual Brain
 - <https://www.thevirtualbrain.org/tvb/zwei/brainsimulator-software>
- Brain Dynamics Toolbox
 - <https://bdttoolbox.org>
- BRIAN Simulator → spiking networks (and more)
 - <https://briansimulator.org/>
- NEURON → Detailed neurons, spiking neurons, networks.
 - www.neuron.yale.edu
- NETPYNE → NETworks in PYthon + NEuron
- NEST → Spiking neurons, networks, detailed neuronal dynamics
 - www.nest-simulator.org
- DIY in Python or Matlab!!

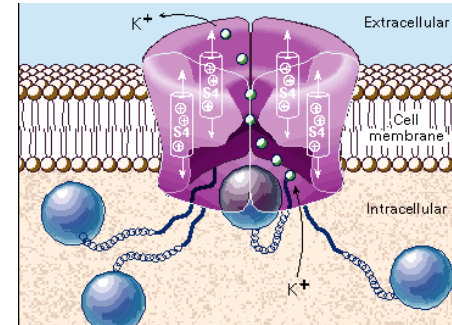
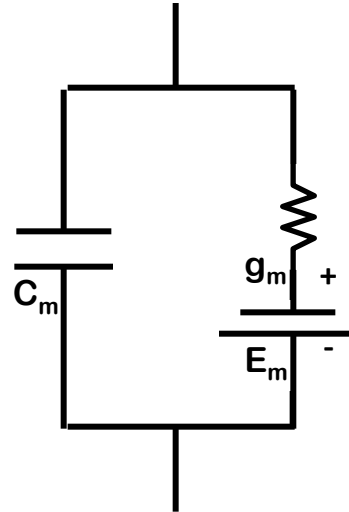
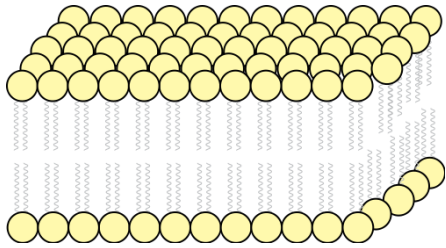
Simulation of neuronal dynamics

Neuronal excitability



Modelando la excitabilidad

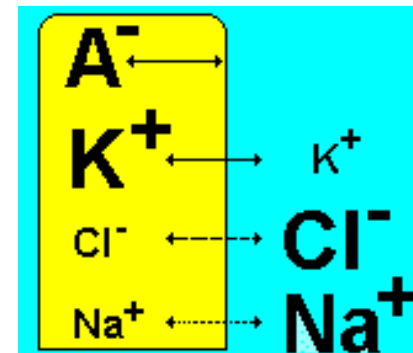
Circuito Equivalente



$$I_{memb} = I_C + I_{ion}$$

$$I_{memb} = C_m \frac{dV}{dt} + g_m (V - E_m)$$

$$C_m \frac{dV}{dt} = -g_m (V - E_m) + I_{ext}$$



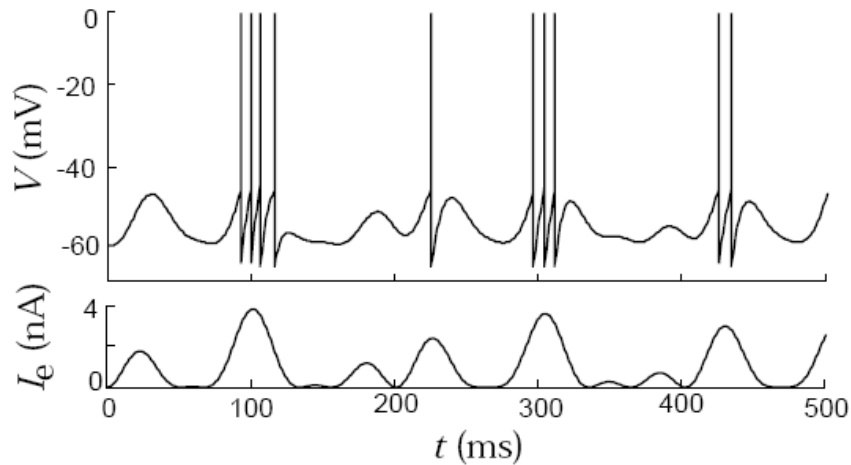
Action potentials?

Integrate-and-fire neurons

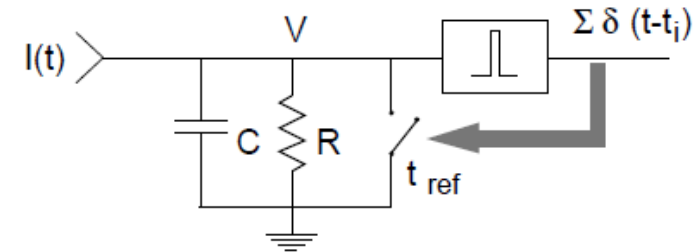
$$C_m \frac{dV}{dt} = -g_m(V - E_m) + I_{ext}(t)$$

$$V(t) \geq V_{th} \Rightarrow V_{(t+1)} = V_{reset}$$

Non linear



Leaky Integrate-and-Fire Unit

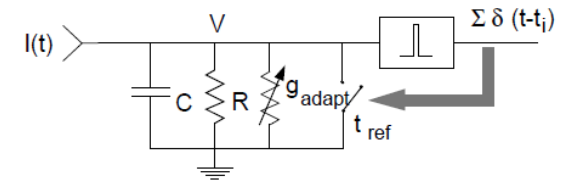


LIF with adaptation

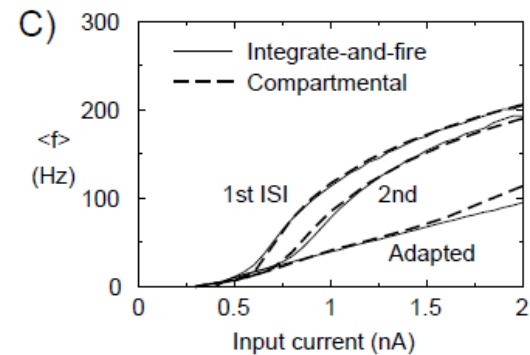
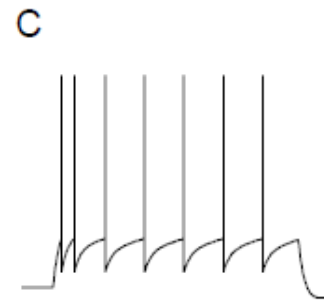
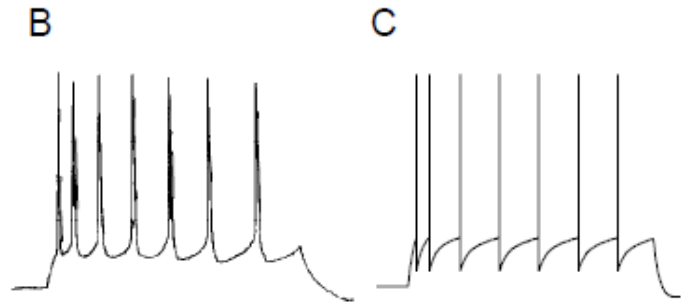
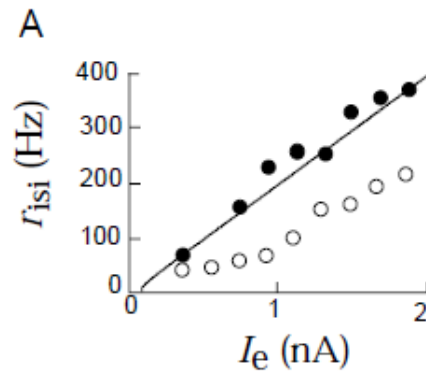
$$C_m \frac{dV}{dt} = -g_m(V - E_m) - g_{adap}(V - E_{adap}) + I_{ext}(t)$$

$$\frac{dg_{adap}}{dt} = -\frac{g_{adap}}{\tau_{adap}}$$

Adapting Integrate-and-Fire Unit



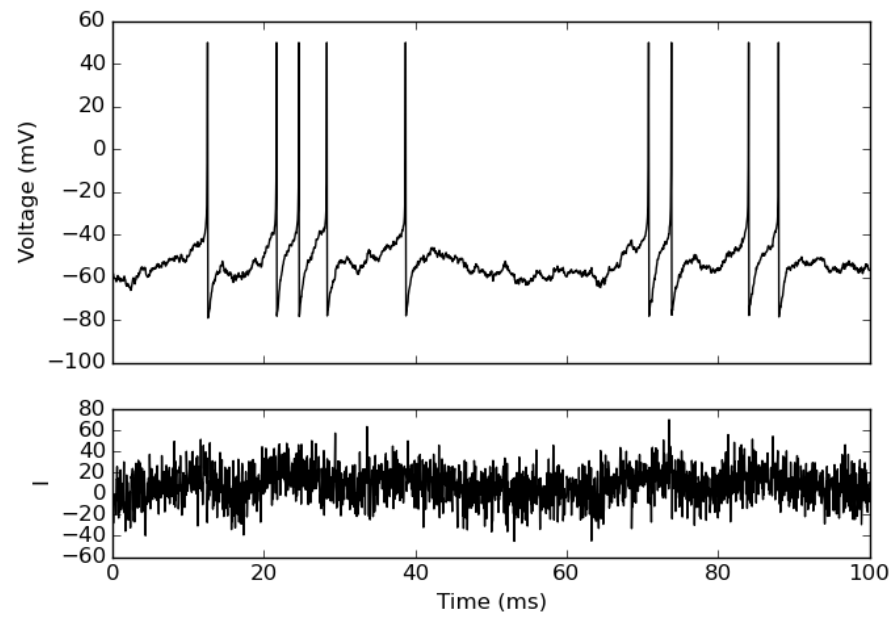
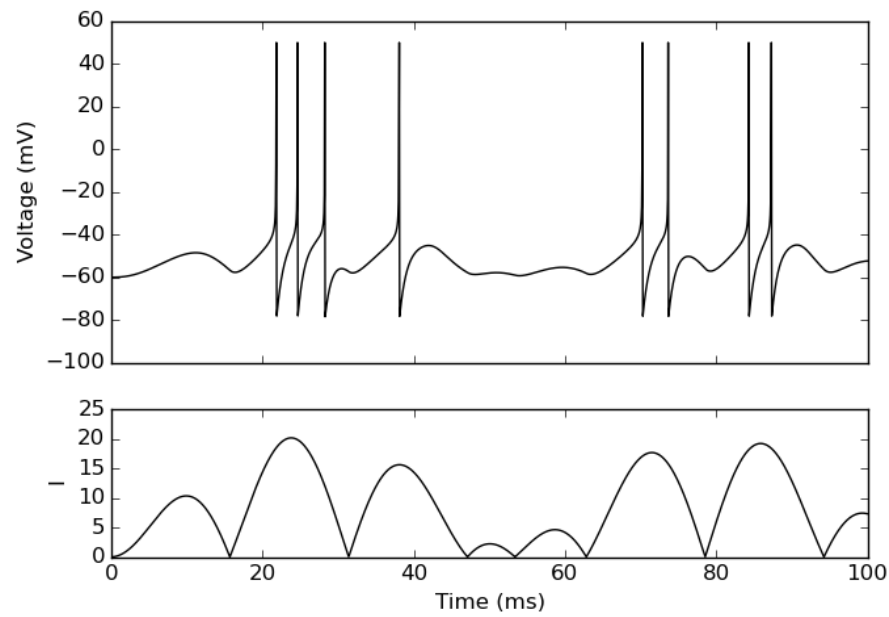
$$V(t) \geq V_{th} \Rightarrow V(t+1) = V_{reset}; g_{adap}(t+1) = g_{adap}(t) + \Delta$$



Exponential Integrate and Fire

$$C \frac{dV}{dt} = -g(V - E) + \Delta T e^{\frac{V - V_{th}}{\Delta T}} + I(t)$$

$$\tau_m \frac{dV}{dt} = -(V - V_{rest}) + \Delta T e^{\frac{V - V_{th}}{\Delta T}} + R_m I(t) \quad V_{(t)} \geq V_{spike} \Rightarrow V_{(t+1)} = V_{reset}$$



Adaptive Exponential Integrate and Fire

- Adaptation in general:

$$\tau_m \frac{dV}{dt} = f(V) - R \sum_k w_k + RI(t)$$

$$\tau_k \frac{dw_k}{dt} = a_k(V - V_{rest}) - w_k + b_k \tau_k \sum_{t^{(f)}} \delta(t - t^{(f)})$$

- AdEx:

$$\tau_m \frac{dV}{dt} = -(V - V_{rest}) + \Delta T e^{\frac{V - V_{th}}{\Delta T}} - R w + RI(t)$$

$$\tau_w \frac{dw}{dt} = a(V - V_{rest}) - w + b \tau_w \sum_{t^{(f)}} \delta(t - t^{(f)})$$

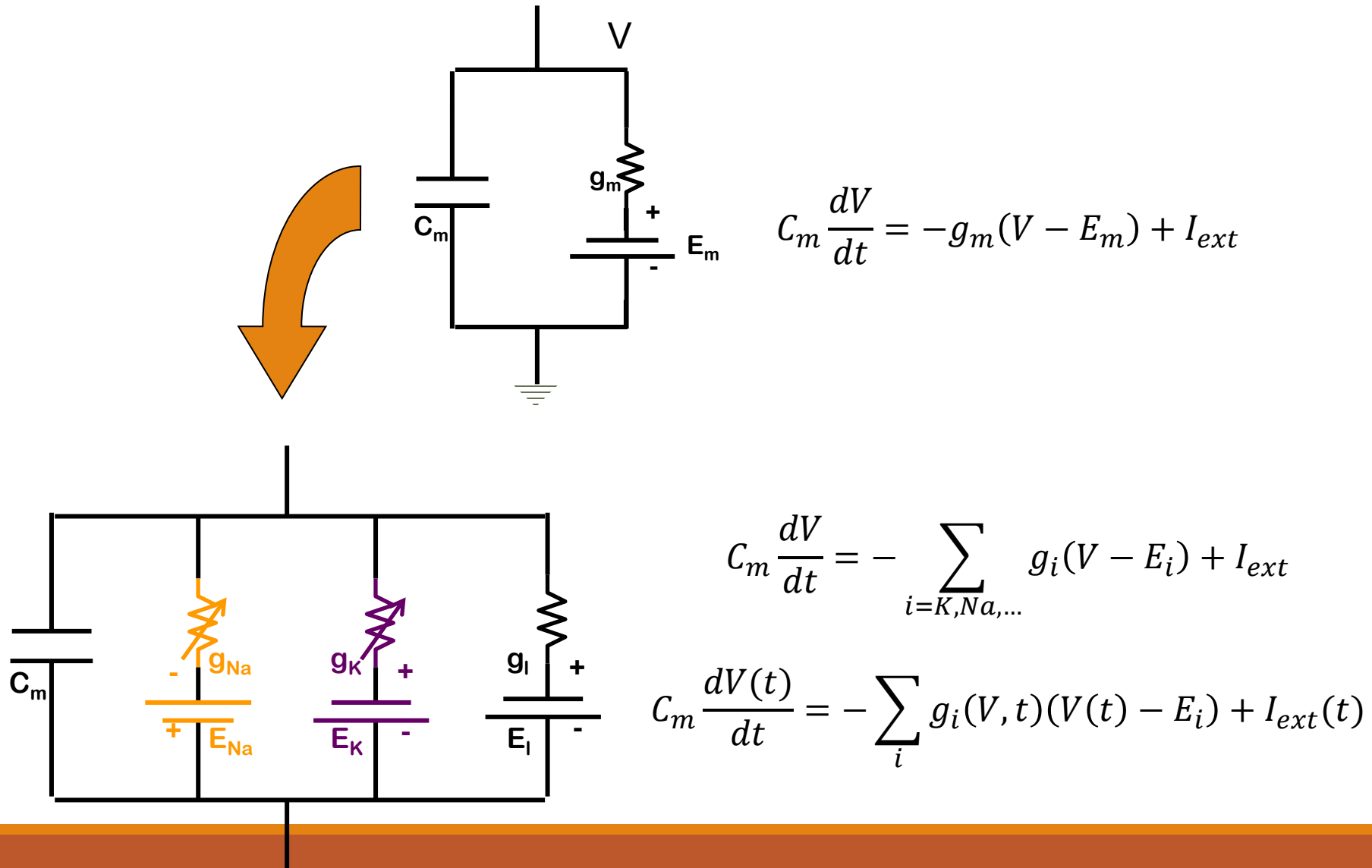
Potenciales de acción

Generados artificialmente cuando V_m llega a un umbral o lo supera.

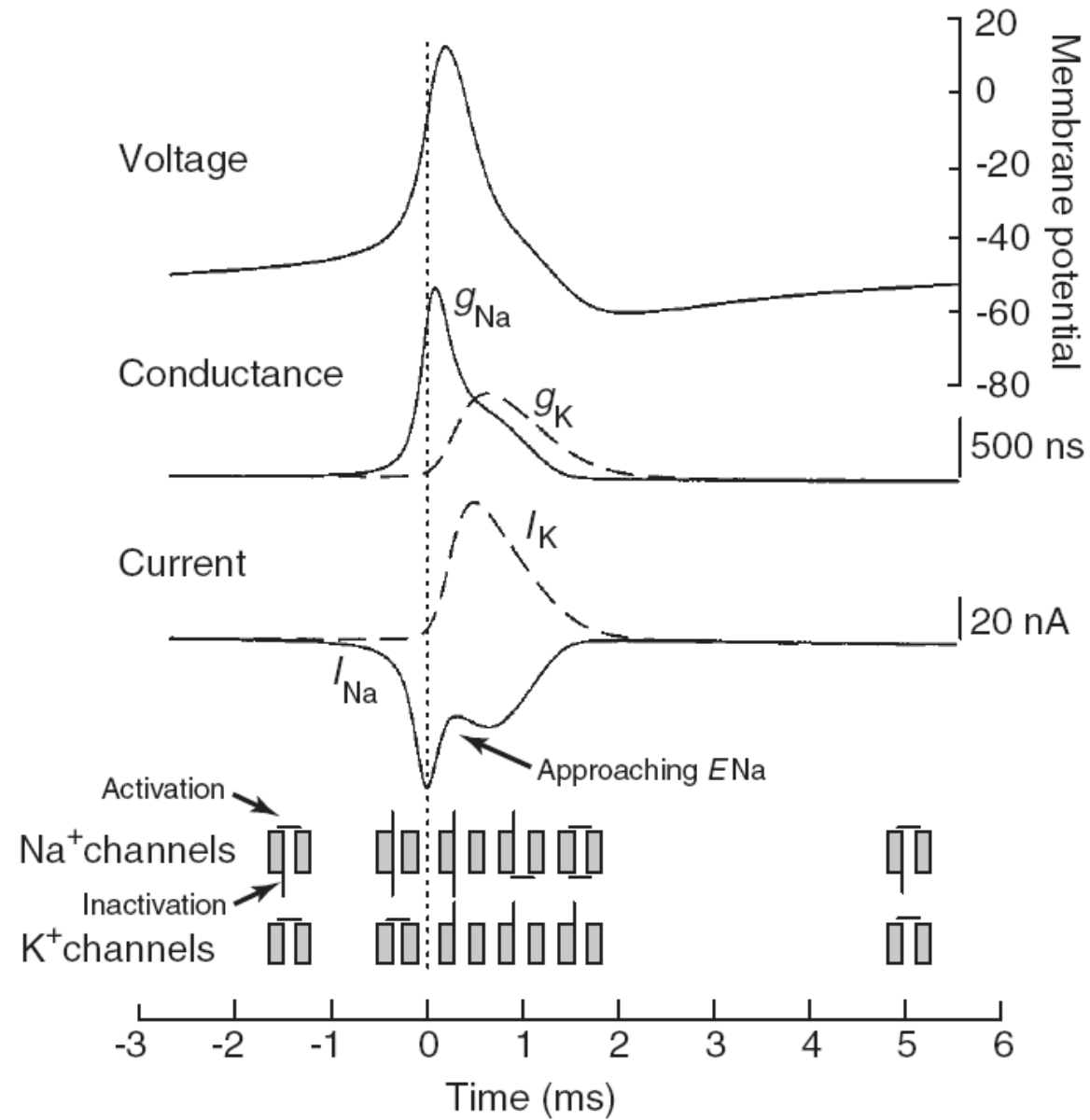
- Neurona '*Integrate-and-fire*' (IAF)

Generados por mecanismos que reflejan el comportamiento de canales de Na^+ y K^+ (u otros) activados por voltaje.

Adding 'realistic' ion channels



Potencial de acción real



Ecuaciones de Hodgkin y Huxley

$$I = C_M \frac{dV}{dt} + \bar{g}_K n^4 (V - V_K) + \bar{g}_{Na} m^3 h (V - V_{Na}) + \bar{g}_l (V - V_l),$$

$$\frac{dn}{dt} = \alpha_n(1 - n) - \beta_n n,$$

$$\frac{dm}{dt} = \alpha_m(1 - m) - \beta_m m,$$

$$C_M \frac{dV}{dt} = f(V, m, h, n)$$

$$\frac{dh}{dt} = \alpha_h(1 - h) - \beta_h h,$$

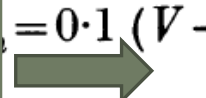
$$\frac{dn}{dt} = g(V, n)$$



Canales de potasio
dependientes de voltaje

$$\frac{dm}{dt} = j(V, m)$$

$$\frac{dh}{dt} = k(V, h)$$



Canales de sodio
dependientes de voltaje

$$\beta_n = 0.1 \exp\left(\frac{V + 30}{10}\right)$$

$$\alpha_m = 0.1 \exp\left(\frac{V + 30}{10}\right)$$

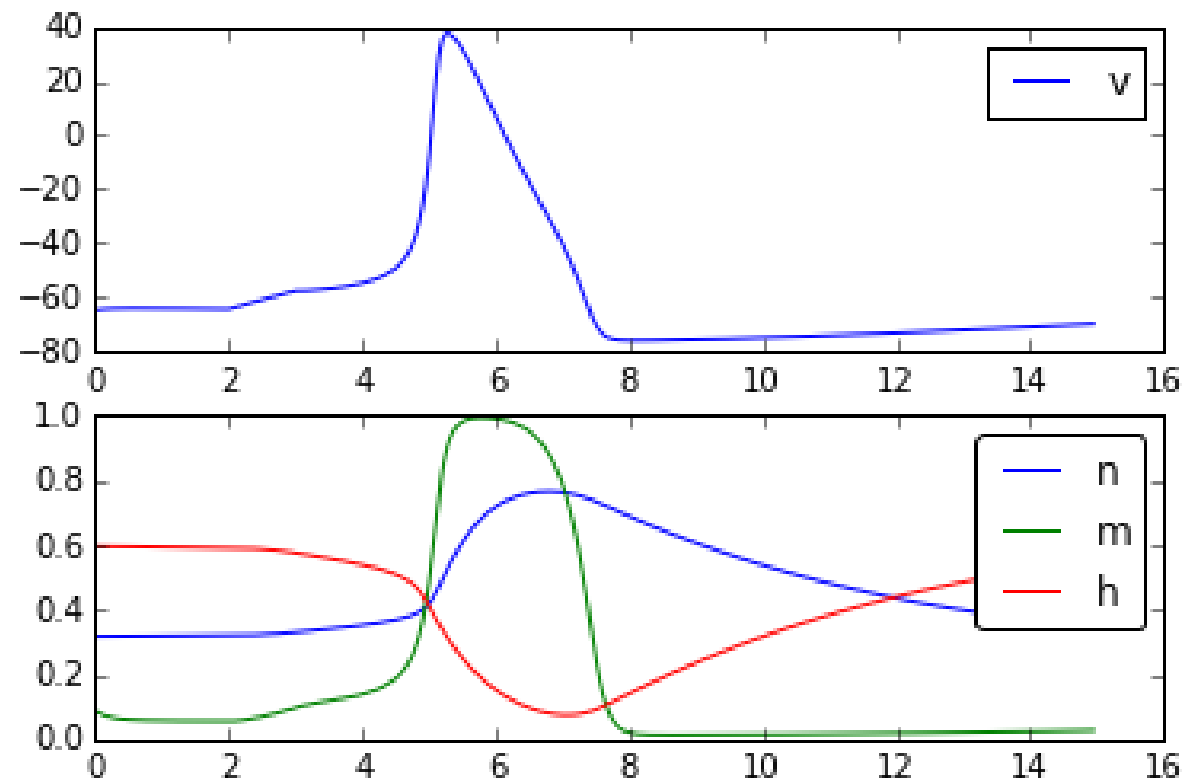
$$\beta_m = 4$$

$$\alpha_h = 0.07 \exp\left(\frac{V + 30}{10}\right)$$

$$\beta_h = 1 / \left(\exp\left(\frac{V + 30}{10}\right) + 1 \right).$$

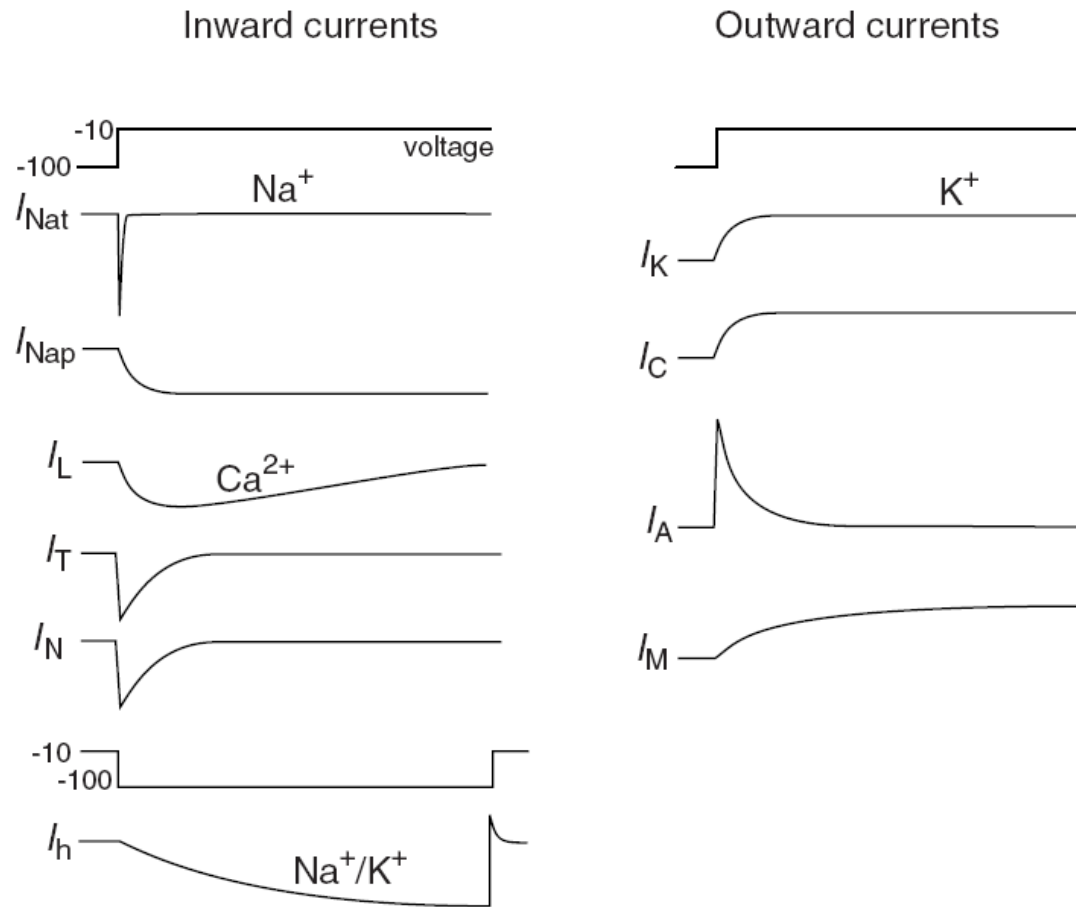
Hodgkin A.L., and Huxley A.F. (1952). A quantitative description of membrane current and its application to conduction and excitation in nerve. *J. Physiol.* **117**: 500-544

Modelo de H&H

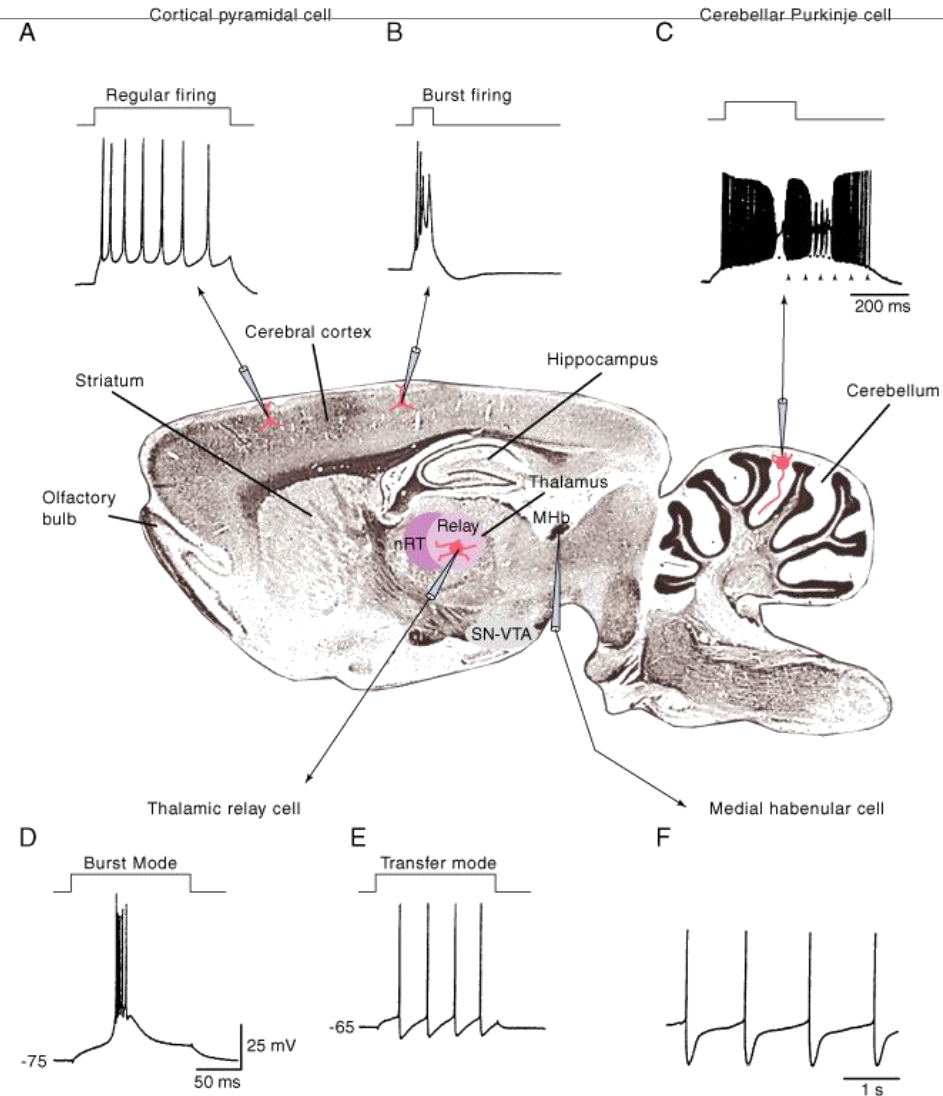


Voltage-dependent ion channels

La respuesta al voltaje no es instantánea



Neuronal
diversity =
ion channel
diversity



A- and C-Type Rat Nodose Sensory Neurons: Model Interpretations of Dynamic Discharge Characteristics

J. H. SCHILD, J. W. CLARK, M. HAY, D. MENDELOWITZ, M. C. ANDRESEN, AND D. L. KUNZE

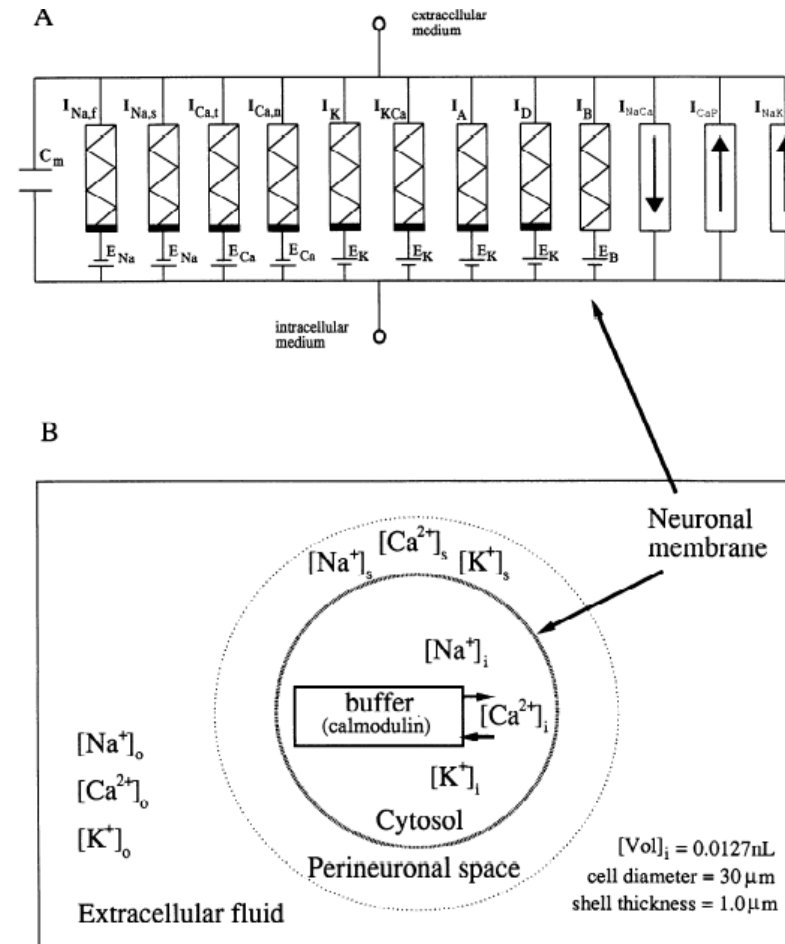


TABLE 1. Inward currents

$I_{Ca,t}$: Low Threshold, Transient Calcium Current

$I_{Ca,t} = \bar{g}_{Ca,t} d_t f_t (V - E_{Ca})$

$\dot{d}_t = \frac{d_{t\infty} - d_t}{\tau_{dt}}$

$\dot{f}_t = \frac{f_{t\infty} - f_t}{\tau_{ft}}$

$E_{Ca} = \frac{RT}{Z_{Ca}F} \ln \frac{[Ca^{2+}]_e}{[Ca^{2+}]_i} - 78.7$

$\tau_{dt} = 22.0 \exp(-(0.052)^2 (V + 68.0)^2) + 2.5$

$\tau_{ft} = 103.0 \exp(-(0.050)^2 (V + 58.0)^2) + 12.5$

$d_{t\infty} = \frac{1.0}{1.0 + \exp((V + 54.00)/-5.75)}$

$f_{t\infty} = \frac{1.0}{1.0 + \exp((V + 68.00)/6.0)}$

$I_{Ca,n}$: High Threshold, Long-Lasting Calcium Current

$I_{Ca,n} = \bar{g}_{Ca,n} d_n (0.55 f_{n1} + 0.45 f_{n2}) (V - E_{Ca})$

$\dot{d}_n = \frac{d_{n\infty} - d_n}{\tau_{dn}}$

$\dot{f}_{n1} = \frac{f_{n1\infty} - f_{n1}}{\tau_{fn1}}$

$\dot{f}_{n2} = \frac{f_{n2\infty} - f_{n2}}{\tau_{fn2}}$

$\tau_{dn} = 3.25 \exp(-(0.042)^2 (V + 31.0)^2) + 0.395$

$\tau_{fn1} = 33.5 \exp(-(0.0395)^2 (V + 30.0)^2) + 5.0$

$\tau_{fn2} = 225.0 \exp(-(0.0275)^2 (V + 40.0)^2) + 75.00$

$d_{n\infty} = \frac{1.0}{1.0 + \exp((V + 20.0)/-4.5)}$

$f_{n1\infty} = \frac{1.0}{1.0 + \exp((V + 20.0)/25.0)}$

$f_{n2\infty} = r_n + \frac{1.0}{1.0 + \exp((V + 40.0)/10.0)}$

$r_n = \frac{0.2}{1.0 + \exp(\frac{V + 5.0}{-10.0})}$

I_{Na_f} : Fast, TTX Sensitive Sodium Current

$I_{Na_f} = \bar{g}_{Na_f} m_f^3 h_f j (V - E_{Na})$

$\dot{m}_f = \frac{m_{f\infty} - m_f}{\tau_{mf}}$

$\dot{h}_f = \frac{h_{f\infty} - h_f}{\tau_{hf}}$

$\dot{j} = \frac{j_{\infty} - j}{\tau_j}$

$\tau_{mf} = 0.75 \exp(-(0.0635)^2 (V + 40.35)^2) + 0.12$

$\tau_{hf} = 6.5 \exp(-(0.0295)^2 (V + 75.00)^2) + 0.55$

$m_{f\infty} = \frac{1.0}{1.0 + \exp((V + 41.35)/-4.75)}$

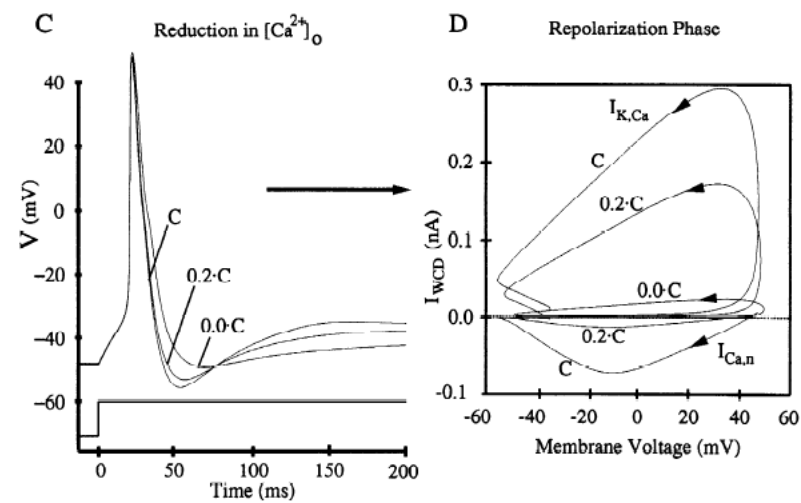
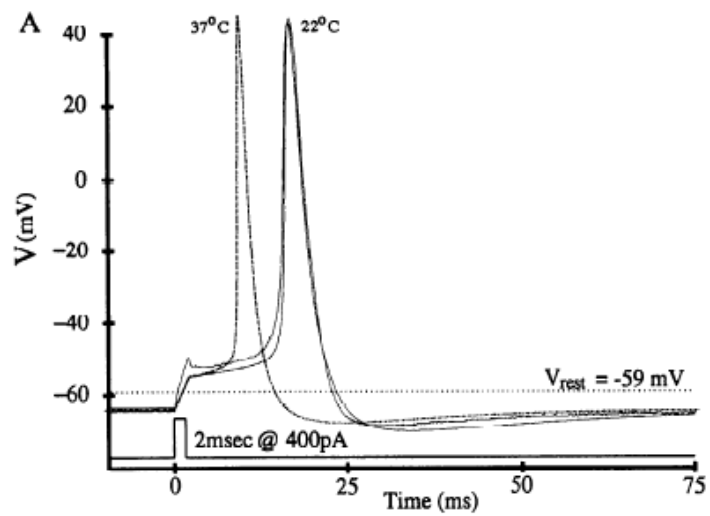
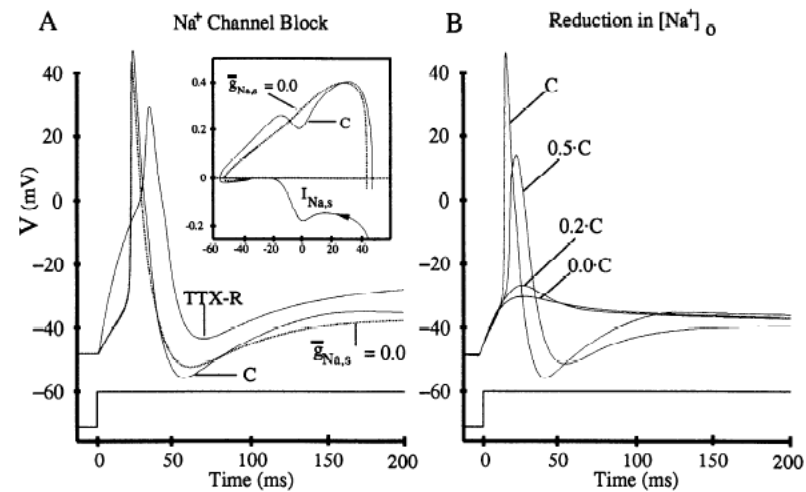
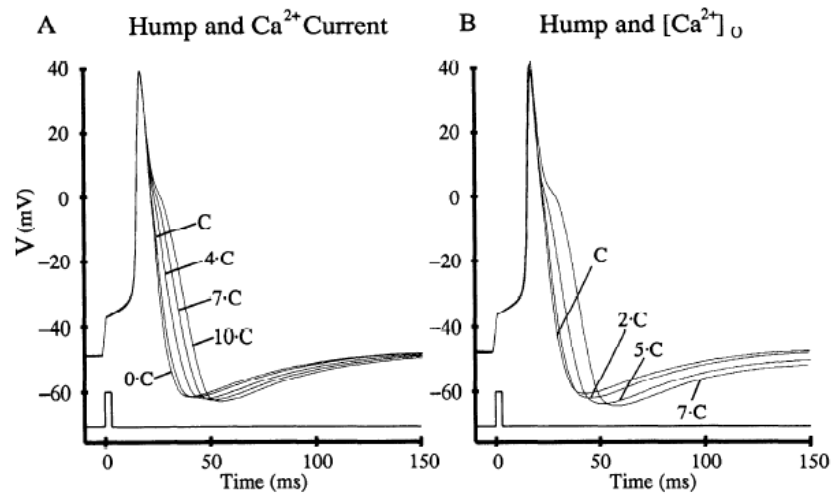
$h_{f\infty} = \frac{1.0}{1.0 + \exp((V + 62.00)/4.50)}$

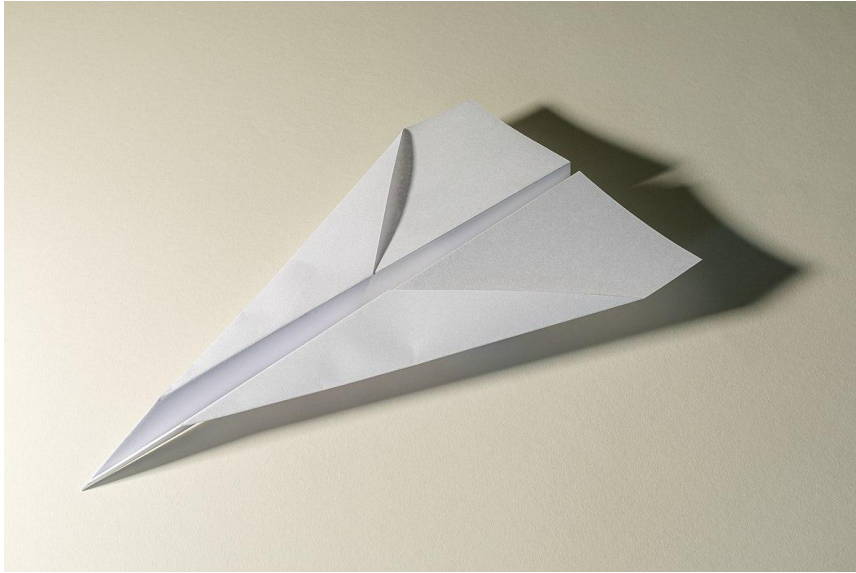
$j_{\infty} = \frac{1.0}{1.0 + \exp((V + 40.00)/1.50)}$

$\tau_j = \frac{25.0}{1.0 + \exp((V - 20.00)/4.50)} + 0.01$

TABLE 2. Outward currents

I_K : Delayed Rectifier	
$I_K = \bar{g}_K n (V - E_K)$	$\dot{n} = \frac{n_{\infty} - n}{\tau_n}$
$\tau_n = \frac{1.0}{(\alpha_n + \beta_n)} + 1.0$	$n_{\infty} = \frac{1.0}{1.0 + \exp((V + 14.62)/-18.38)}$
$\alpha_n = \frac{0.001265(V + 14.273)}{1.0 - \exp((V + 14.273)/-10.0)}$	$\beta_n = 0.125 \exp(\frac{V + 55.0}{-2.5})$
I_A : Early Transient Outward Current	
$I_A = \bar{g}_A p^3 q (V - E_K)$	
$\dot{p} = \frac{p_{\infty} - p}{\tau_p}$	$\dot{q} = \frac{q_{\infty} - q}{\tau_q}$
$\tau_p = 5.0 \exp(-(0.022)^2 (V + 65.0)^2) + 2.5$	$\tau_q = 100.0 \exp(-(0.035)^2 (V + 30.0)^2) + 10.5$
$p_{\infty} = \frac{1.0}{1.0 + \exp((V + 28.0)/-28.0)}$	$q_{\infty} = \frac{1.0}{1.0 + \exp((V + 58.0)/7.0)}$
I_D : Slowly Inactivating Delay Current	
$I_D = \bar{g}_D x^3 y (V - E_K)$	
$\dot{x} = \frac{x_{\infty} - x}{\tau_x}$	$\dot{y} = \frac{y_{\infty} - y}{\tau_y}$
$\tau_x = 5.0 \exp(-(0.022)^2 (V + 65.0)^2) + 2.5$	$\tau_y = 7500.0$
$x_{\infty} = \frac{1.0}{1.0 + \exp((V + 39.59)/-14.68)}$	$y_{\infty} = \frac{1.0}{1.0 + \exp((V + 48.0)/7.0)}$
$I_{K,Ca}$: Calcium-Activated Potassium Current	
$I_{K,Ca} = \bar{g}_{K,Ca} c (V - E_K)$	$\dot{c} = \frac{c_{\infty} - c}{\tau_c}$

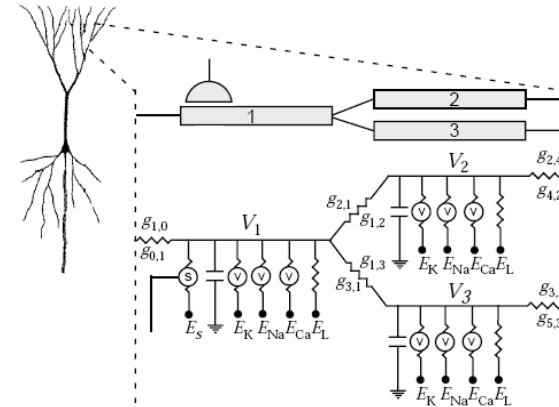
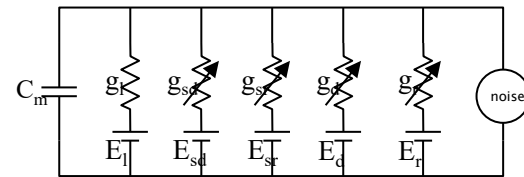
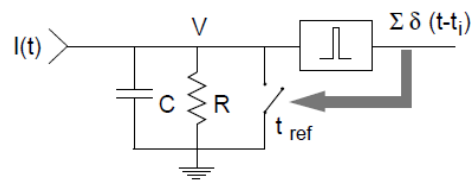
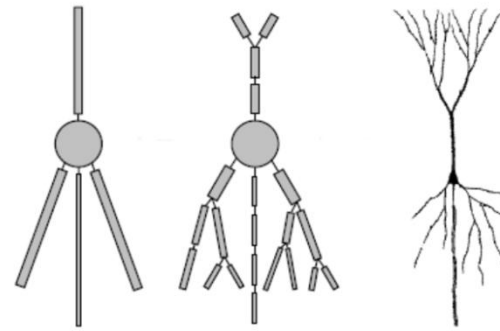




What is the most realistic model?

DO WE WANT TO BE REALISTIC?





Modelos tipo IF

Modelos basados
en conductancias
(HH)

Modelos multi-
compartimento
(geometría)

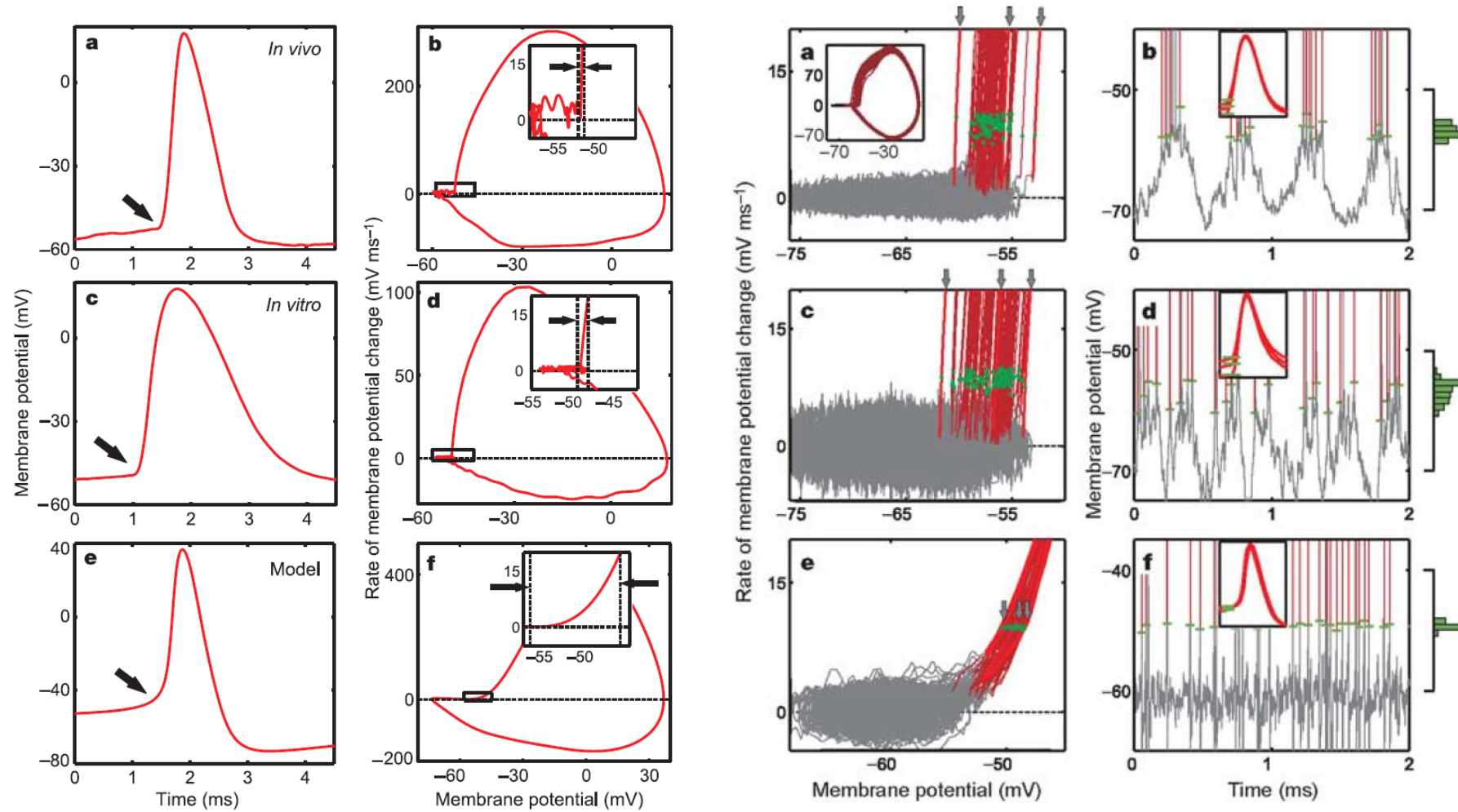
Complejidad

Apego a la realidad

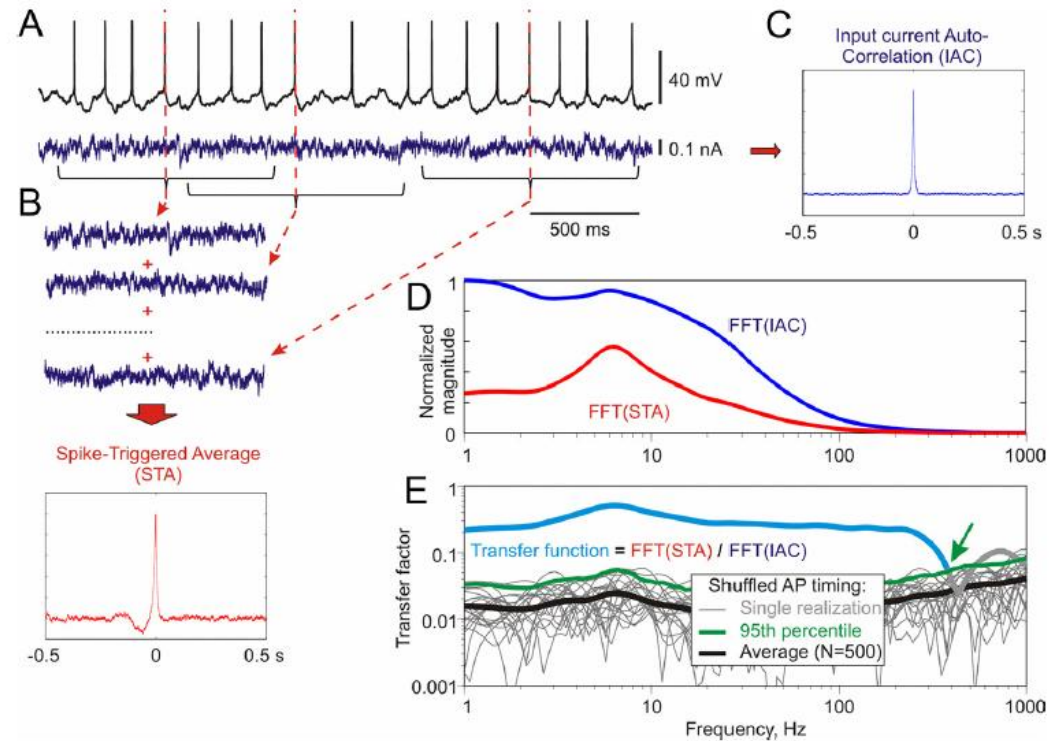
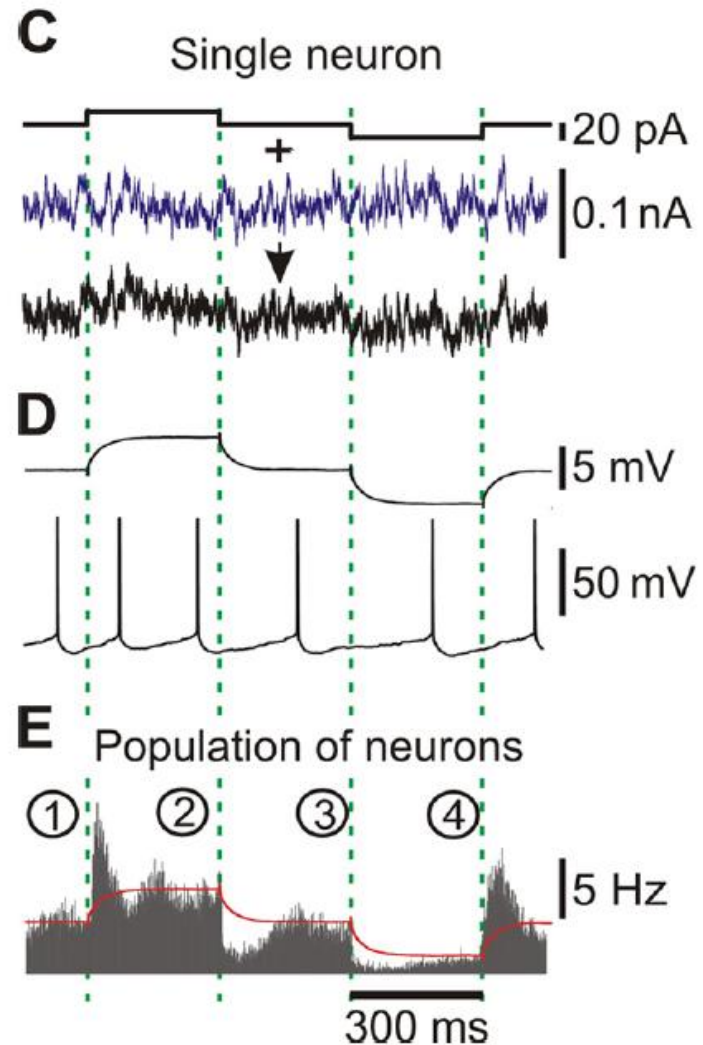
- Numérica
- Matemática



Fast onset of the action potential

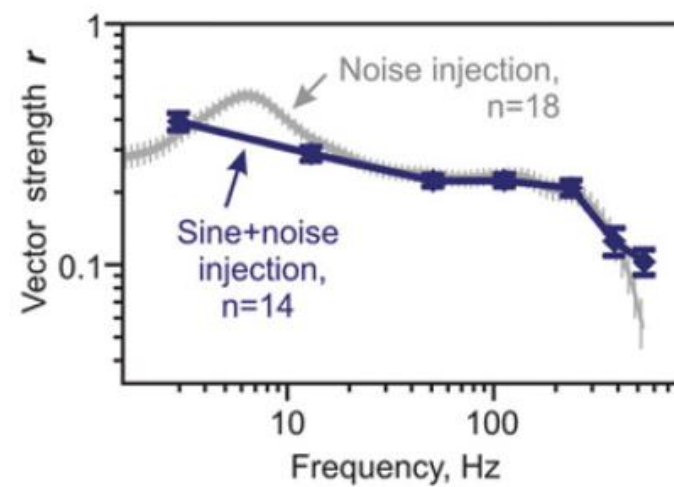
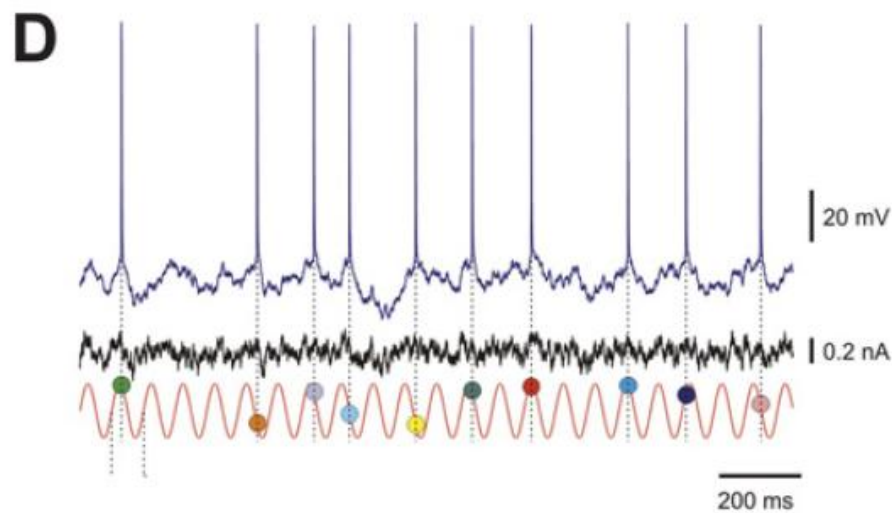
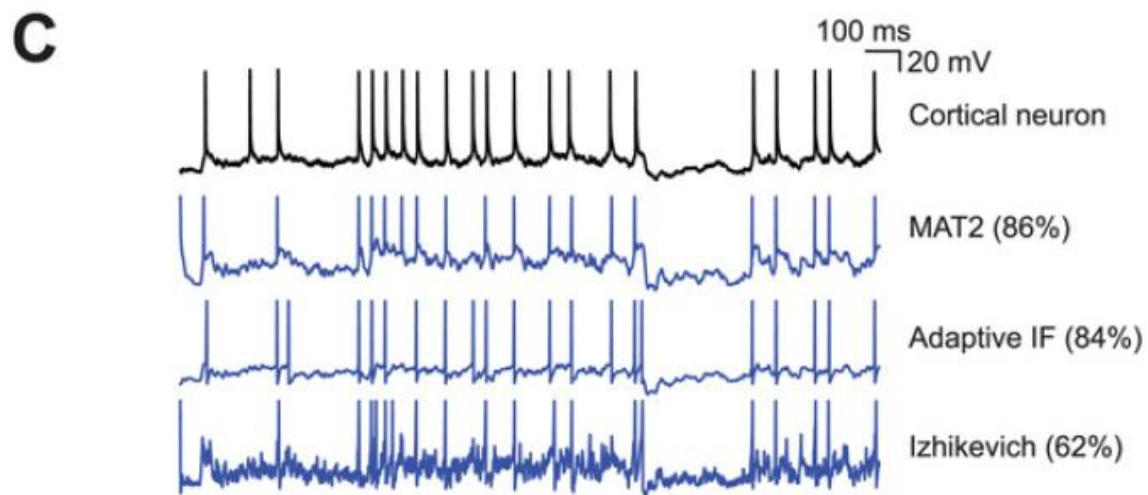


High-frequency codification



Tchumatchenko *et al.* (2011) *J Neurosci* **31**:12171–12179

Ilin *et al.* (2013) *J Neurosci* **33**: 2281–2292



Hypotheses

Sodium channel cooperativity

Backpropagation 'artifact'

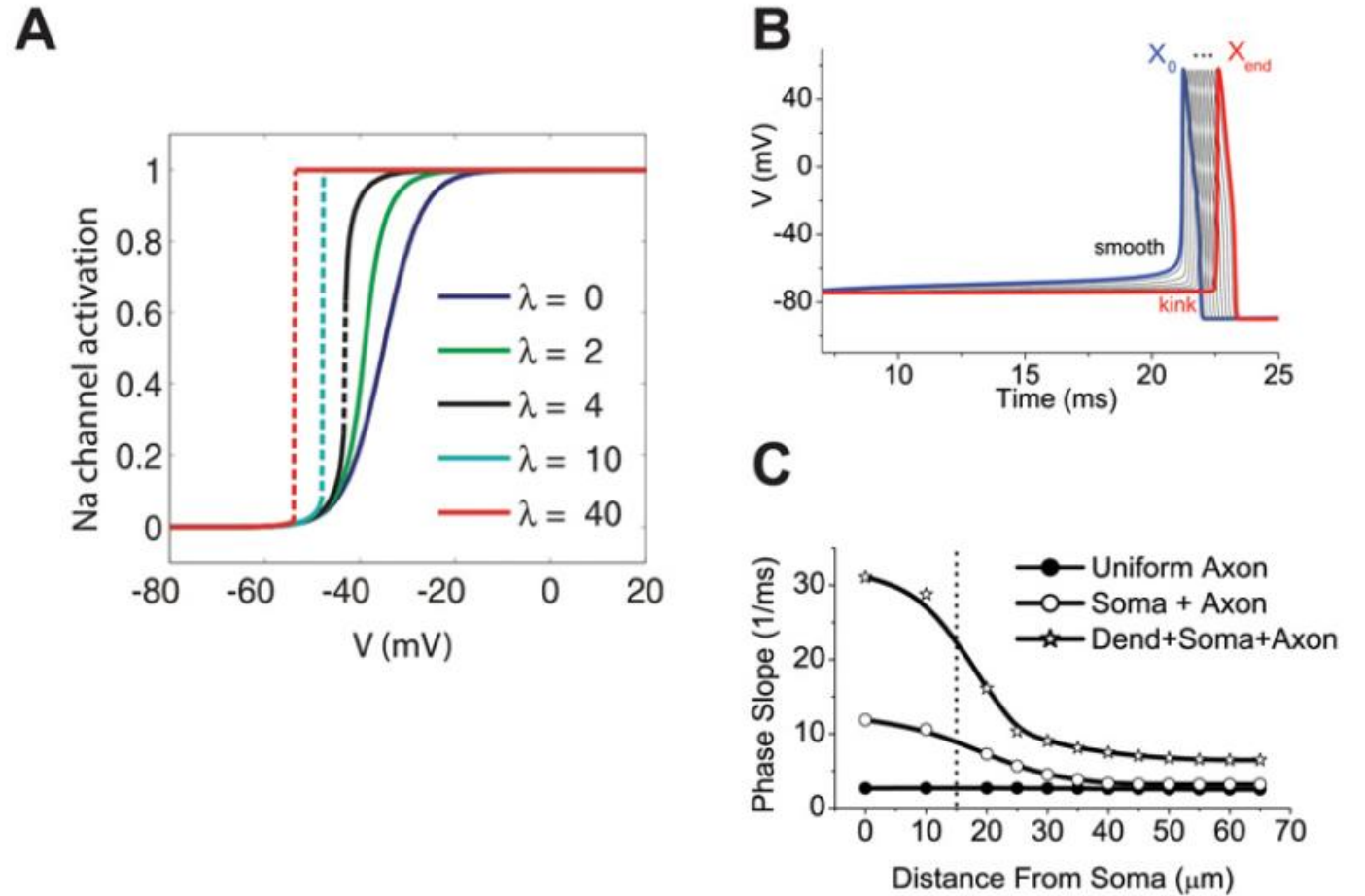
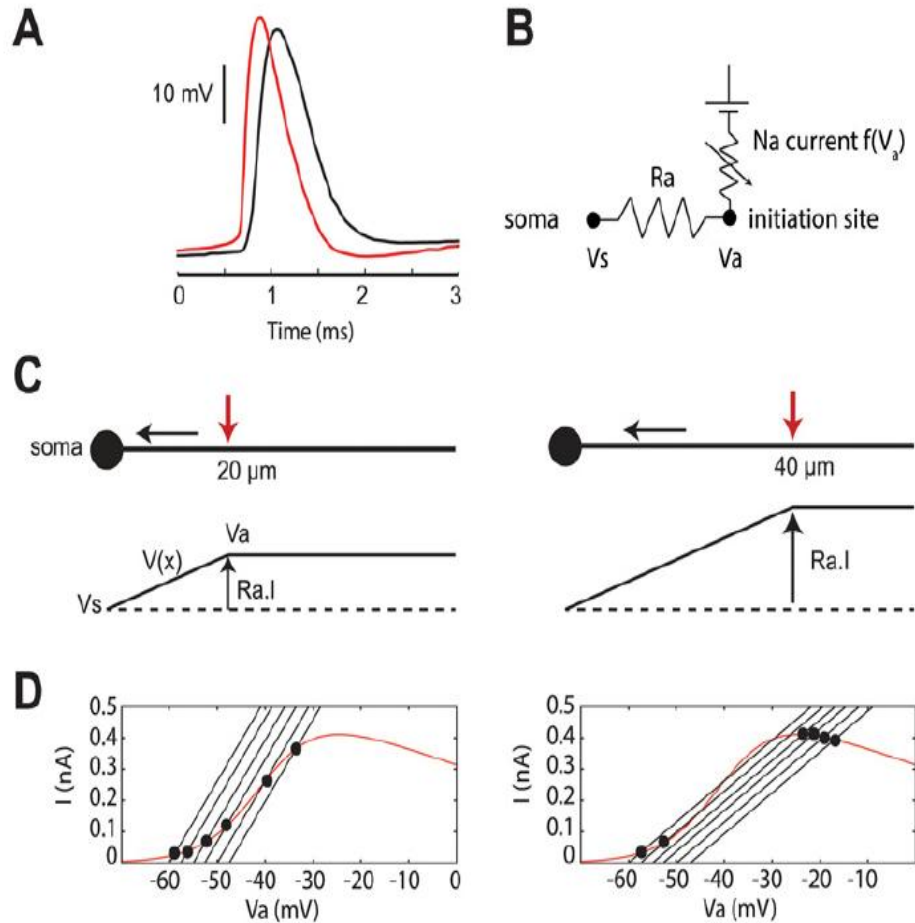


Fig 4. Cooperativity and backpropagation hypotheses. (A) The cooperativity hypothesis (adapted from

An explanation based of the geometry of AP initiation site



Análisis matemático de la propagación de corrientes soma $\leftrightarrow$ AIS

Brette R., *PLOS Comp Biol* 2013

Brette R., *PLOS Comp Biol* 2015

Modelos
basados en
conductancias



Modelos tipo
IF



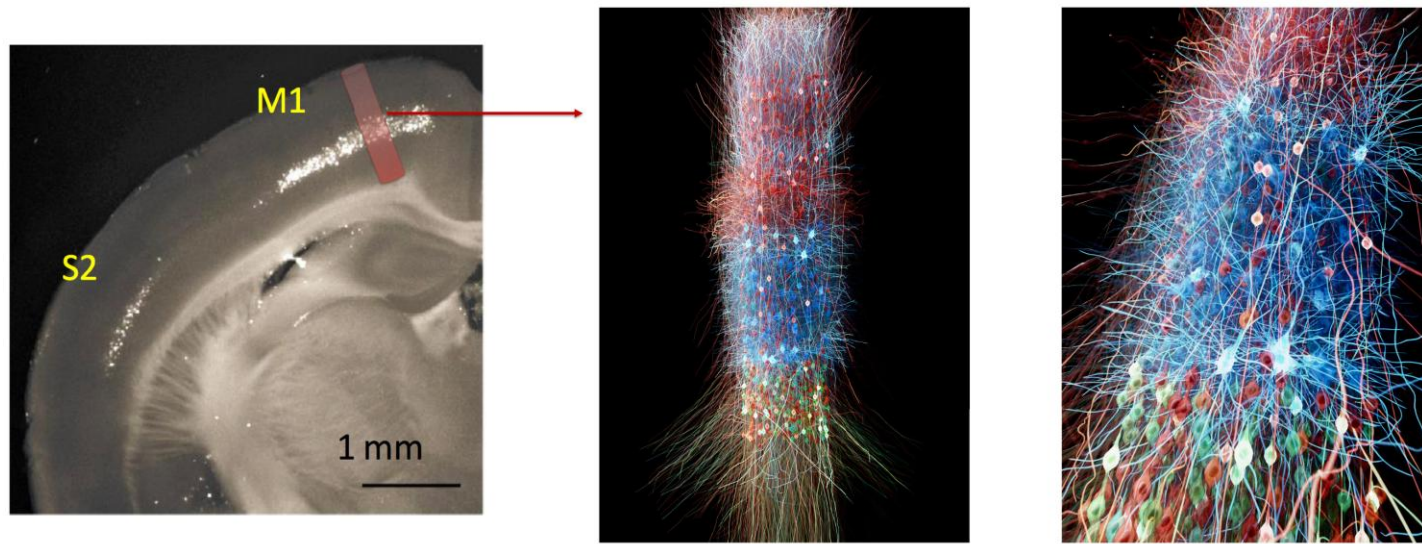
Modelos multi-
compartimento
(geometría)

Apego a la realidad

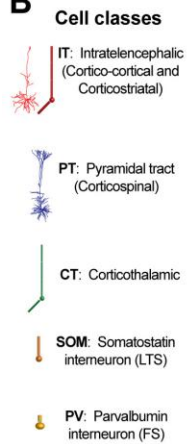
Complejidad
(tiempo de cómputo)



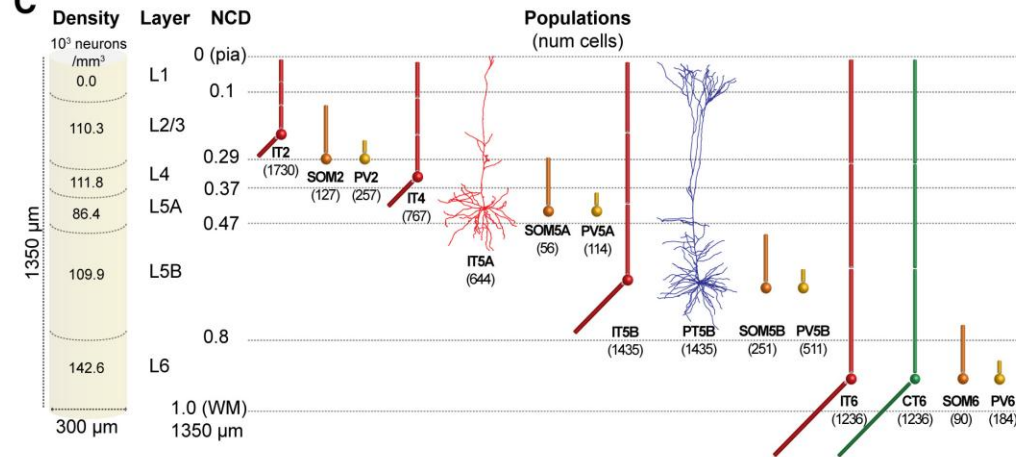
A



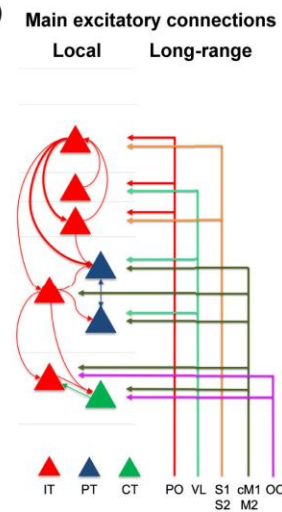
B



C



D



Dura-Bernal, S., Neymotin, S. A., Suter, B. A., Dacre, J., Moreira, J. et al., (2023). Multiscale model of primary motor cortex circuits predicts in vivo cell-type-specific, behavioral state-dependent dynamics. *Cell Reports*.



**"All models are wrong,
but some are useful"**
George E.P. Box

Del rigor en la ciencia – Jorge Luis Borges

En aquel Imperio, el Arte de la Cartografía logró tal Perfección que el mapa de una sola Provincia ocupaba toda una Ciudad, y el mapa del Imperio, toda una Provincia. Con el tiempo, estos Mapas Desmesurados no satisficieron y los Colegios de Cartógrafos levantaron un Mapa del Imperio, que tenía el tamaño del Imperio y coincidía puntualmente con él.

Menos Adictas al Estudio de la Cartografía, las Generaciones Sigüientes entendieron que ese dilatado Mapa era Inútil y no sin Impiedad lo entregaron a las Inclemencias del Sol y los Inviernos. En los desiertos del Oeste perduran despedazadas Ruinas del Mapa, habitadas por Animales y por Mendigos; en todo el País no hay otra reliquia de las Disciplinas Geográficas.

Suárez Miranda, Viajes de Varones Prudentes, Libro Cuarto, Cap. XLV, Lérida, 1658.

On Exactitude in Science

Jorge Luis Borges, *Collected Fictions*, translated by Andrew Hurley.

...In that Empire, the Art of Cartography attained such Perfection that the map of a single Province occupied the entirety of a City, and the map of the Empire, the entirety of a Province. In time, those Unconscionable Maps no longer satisfied, and the Cartographers Guilds struck a Map of the Empire whose size was that of the Empire, and which coincided point for point with it. The following Generations, who were not so fond of the Study of Cartography as their Forebears had been, saw that that vast Map was Useless, and not without some Pitilessness was it, that they delivered it up to the Inclemencies of Sun and Winters. In the Deserts of the West, still today, there are Tattered Ruins of that Map, inhabited by Animals and Beggars; in all the Land there is no other Relic of the Disciplines of Geography.

—Suárez Miranda, *Viajes de varones prudentes*, Libro IV, Cap. XLV, Lérida, 1658